

Radian Measure and Arc Length

ANSWER KEY



Converting between degrees and radians

1 Convert each angle to radians, in exact form:

a $45^\circ = \frac{\pi}{4}$

b $120^\circ = \frac{2\pi}{3}$

c $270^\circ = \frac{3\pi}{2}$

d $330^\circ = \frac{11\pi}{6}$

2 Convert each angle to degrees. Round to one decimal place where needed:

a $\frac{\pi}{6} = 30^\circ$

b $\frac{5\pi}{4} = 225^\circ$

c $\frac{7\pi}{3} = 420^\circ$

d $3.5 \text{ rad} \approx 200.5^\circ$

Arc length and sector area

3 A circle has radius 6 cm. Find the exact length of the arc subtended by a central angle of $\frac{2\pi}{3}$.

$$s = r\theta = 6 \cdot \frac{2\pi}{3} = 4\pi \approx 12.57 \text{ cm.}$$

4 A sector of a circle of radius 10 cm has central angle $\frac{\pi}{5}$. Find its exact area.

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\frac{\pi}{5} = 10\pi \approx 31.42 \text{ cm}^2.$$

5 A wheel of radius 30 cm rolls along the ground without slipping.

a Through what angle, in radians, does the wheel turn when it travels 1.5 m? $\theta = \frac{s}{r} = \frac{150}{30} = 5$ radians.

b Express this angle to the nearest tenth of a degree. $5 \times \frac{180^\circ}{\pi} \approx 286.5^\circ$.

6 A pendulum of length 80 cm swings through an angle of 25° . Find the length of the arc traced by its tip, to the nearest tenth of a centimetre.

$$\theta = 25^\circ = \frac{25\pi}{180} \text{ rad, so } s = 80 \cdot \frac{25\pi}{180} \approx 34.9 \text{ cm.}$$